



Compatibility of Host Plant Resistance and biological control



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Abstract

Host plant resistance can be defined as the relative amount of heritable qualities possessed by the plant that influences the ultimate degree of damage done by the insect. HPR and biological control can be defined as key components of integrated pest management, but their interactions are complex. Various plant defensive traits such as semiochemicals, toxins, nutrients, physical structures can directly and indirectly impact natural enemies, enhancing or disrupting their efficiency. Therefore, Optimising pest management requires employing intermediate resistance levels that are compatible with natural enemies. Breeding should aim for this synergy, fostering an intermediate level of resistance and conservation of beneficials. This article talks about the compatibility of HPR with biological control, with relevant examples.

Key words: Host Plant Resistance, Biological Control, Natural enemies, Compatibility, Semiochemicals and Trichomes.

Introduction:

A wide range of defence mechanisms is employed by crops against pests and diseases (Stout *et al.*, 2014). These defensive traits may be naturally occurring or artificially selected through crop breeding. While these traits can have obvious and direct impacts on herbivorous pests, many have profound effects on higher trophic levels which include the natural enemies. Multi-trophic effects of host plant resistance have the potential to influence biological control both positively and negatively. Some resistance mechanisms that protect plants from herbivorous insects may harm or repel the natural enemies thereby promoting incompatibility between HPR and biological control. Complex interactions between HPR and biological control agent can shape the outcome of an IPM program. Such interactions are rarely explored during plant breeding programs. According to Peterson *et al.* (2016), plant defense traits can influence natural enemies through four primary types of interactions: semiochemical mediation, plant toxins, plant nutrients and physical traits.

Semiochemical mediated interactions between plants and natural enemies

Plants produce volatile compounds that are used by arthropod herbivores to locate suitable host plants (Schoonhoven *et al.*, 2005). These volatile profiles can change both quantitatively and qualitatively following herbivory, thus altering their attractiveness (or repellency) to herbivores and their natural enemies (Heil, 2014). Feeding by chewing insects causes mechanical damage to plant tissues, thereby eliciting a wound response. This creates electrical, hydraulic and chemical signals (e.g., systemin; Kessler and Baldwin, 2002). After signal production as local and systemic release of linoleic acid takes place from the plant cell membrane. Linolenic acid is converted into 13-hydroperoxide by lipoxygenase (LOX), which enters one of two pathways (Walling, 2000; Kessler and Baldwin, 2002). In one pathway, 13-hydroperoxide will be hydrolysed by hydroperoxide lyase to yield 'green leaf volatiles' (e.g., C6 alcohols, aldehydes and other volatiles such as terpenoids) thus promoting indirect defenses as they attract natural enemies. Alternatively, is 13-hydroperoxide enters into octadecanoid pathway, it will promote direct defense.

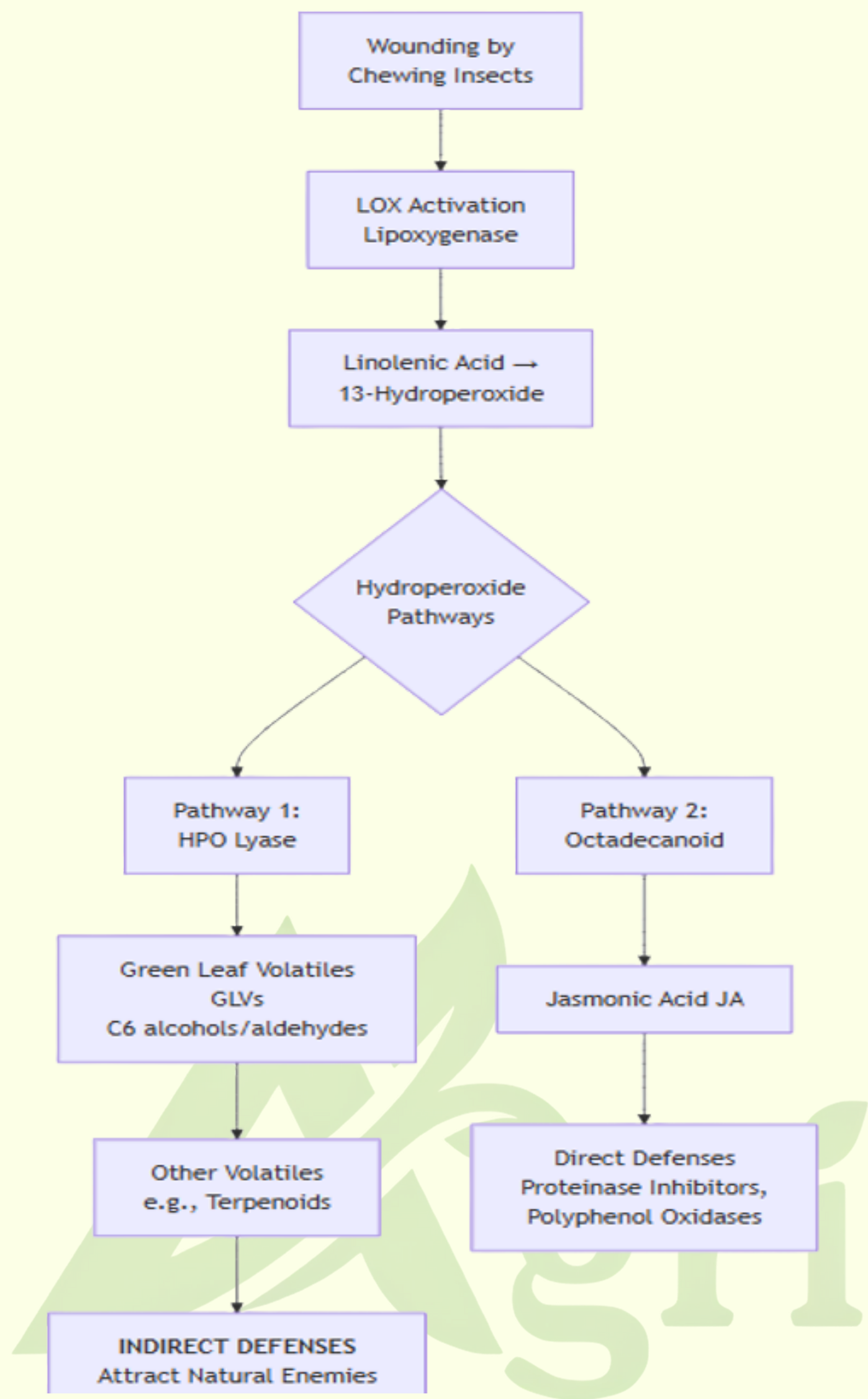


Fig 1. Plant defenses induced by chewing insects influencing natural enemies

Examples:

- Tomato plants attacked by tobacco budworm *Heliothis virescens* but not the closely related tomato fruit worm *Helicoverpa zea*, emit a volatile profile that is highly attractive to the specialist parasitoid of the tobacco budworm, *Cardiochiles nigriceps*. (De Moraes *et al.*, 1998).
- In studies of *Cotesia glomerata* attacking *Pieris rapae*, plants attacked by more herbivores or induced with higher concentrations of JA (simulating higher levels of herbivory) were more attractive to *C. glomerata* (Bruinsma *et al.*, 2009).

Plant toxin mediated interaction and their effect on natural enemies

The plant toxins produced constitutively or induced (as a result of herbivory) may reduce host size thus negatively affecting the parasitoids parasitising on such hosts or can be passed unmetabolized through the herbivore’s midgut into the hemolymph where they are directly encountered by developing parasitoid larvae (Lampert *et al.*, 2008). These toxins can also be sequestered by the insect pest for defense against their own natural enemies (Lampert *et al.*, 2011b).

For example, the catalpa sphinx moth, *Ceratomia catalpa*, sequesters the iridoid glycoside catalpol when it feeds on the *Catalpa bignonioides* Walter (Lampert *et al.*, 2010). The parasitoid of the pest, *Cotesia congregata* accumulates catalpol in the tissues and uses against its own hyperparasitoids (Lampert *et al.*, 2011a). Thus, the interaction of HPR and biological control is multitrophic. Herbivores feeding on a broader range of host plants typically detoxifies a broad array of plant toxins (Ali and Agrawal, 2012). Herbivores with specialized diets can metabolize a narrower range of plant toxins to which they are exposed (Mao *et al.*, 2006). Significantly more xanthotoxin was passed unmetabolized into the hemolymph of *Trichoplusia ni*, a generalist herbivore, than was passed in the hemolymph of the parsnip specialist *Depressaria pastinacella* (Geeze). In turn, *Copidosoma floridanum* (a parasitoid of *T. ni*) suffered increased mortality and reduced clutch sizes relative to *Copidosoma sosares* (a specialist parasitoid of *D. pastinacella* (Lampert *et al.*, 2011a).

Plant nutrient-based interactions and their influence on natural enemies

The nutrients provided by plants to insect herbivores affect the nutrients subsequently available to their natural enemies. The availability, quantity and quality of these nutrients varies depending on biotic and abiotic factors (Fox *et al.*, 1990). When the growth and development of the herbivore is delayed due to suboptimal plant quality, the period of time when herbivores are vulnerable to attack is extended (Moran and Hamilton, 1980). This results in an indirect interaction between host plant nutrition and natural enemies. The slow growth high mortality hypothesis was reported for *Epilachna varivestis* feeding on soybean. The spined soldier bug, *Podisus maculiventris*, was better able to control *E. varivestis* on crop varieties that lowered the herbivore's growth rate. Here, the exact resistance mechanism was not known. *S. frugiperda* feeding on resistant maize plants had reduced growth and vigor. This made them more susceptible to NPV infection (Hamm and Wiseman, 1986). Another example of indirect HPR is the extra floral nectaries which attract predators and parasitoids for the protection of the plant against herbivores. They provide nutrition to the natural enemies such as ants, spiders, predatory mites, coccinellids and parasitoids. EFN can be constitutive and inducible (Heil, 2015).

Physically mediated interactions and their impact on natural enemies

Plant architecture affects the dispersion of herbivores on a host plant, which may in turn affect searching behaviour and host-finding abilities of natural enemies. For example, the leaves of winter wheat varieties with flat leaves were resistant to Russian wheat aphid *Diuraphis noxia*. But in susceptible varieties, leaves furl in response to aphid feeding (Hawley *et al.*, 2003). Thus, flat leaved resistant varieties exposed aphids to disturbances such as wind, rain, and predators. Characteristics that affect falling behavior of herbivores can affect predation rates.

Leaf surface structures that defend the plant from herbivores, such as leaf toughness, cuticle thickness, epicuticular waxes, trichomes and spines, can have direct and indirect effects on natural enemies. An indirect effect can occur if physical defense traits, such as leaf toughness, delay the development of herbivores. In general, trichomes have more harmful than beneficial effects on predators, although most of these effects are sublethal (Riddick and Simmons, 2014). Biological control is possible on glabrous cucumber varieties, however, dense trichomes reduce the searching efficiency by the parasitoid *Encarsia formosa* attacking greenhouse whiteflies *Trialeurodes vaporariorum*. On the contrary, Clusters of trichomes on the underside of plant leaves can form domatia, commonly used by predatory arthropods for shelter.

Host plant resistance conferred by the glossy phenotypes of *Brassica oleracea* is also enhanced by predators. Field studies revealed that green lacewing *Chrysoperla carnea*, insidious flower bug *Orius insidiosus* and convergent lady beetle *Hippodamia convergens*, all generalist predators, significantly increased *P. xylostella* larval mortality in glossy, but not normal wax (glaucous) varieties (Eigenbrode *et al.*, 1995). The reduction in mining behaviour renders the larvae more exposed to predators. Predators also walked faster, spent more time walking and covered more leaf area on glossy leaves compared to normal wax varieties (Eigenbrode *et al.*, 1996). Increased mobility of the natural enemies was attributed to increased traction/adhesion of predators on the glossy phenotype of *B. oleracea*.

Examples on the influence of HPR on Natural enemies

When RWG (Resistant Wild Genotypes), RDG (Resistant Domesticated Genotypes) and SDG (Susceptible Domesticated Genotype) were compared for their influence on *O. laevigatus* in tomato ecosystem by (Guruswamy *et al.*, 2023) it was observed that, tomato genotypes had a significant effect on mortality rates at the early nymphal stage and on total mortality, but not at the late nymphal stage. A significant positive correlation was found between the total glandular trichome density and early nymphal mortality and total immature mortality, and between total trichomes and early nymphal mortality and total immature mortality.

When six tomato genotypes with varying resistance to *P. absoluta* were investigated to analyze the performance of three key natural enemies, the egg parasitoid *Trichogramma achaeae*, the larval parasitoid *Necremnus tuta*, and the predatory mirid *Macrolophus pygmaeus* by (Zannou *et al.*, 2025), it was observed that the resistant wild *Solanum arcanum* negatively affected both parasitoids.

In the research conducted by War *et al.*, 2016, the behavioral responses of two natural enemies of *Helicoverpa armigera* including, *Trichogramma chilonis* and *Campoletis chlorideae* were tested against several groundnut genotypes using a Y-tube olfactometer. It was found that the leaves of resistant genotypes ICGV 86699, ICGV 86031, and ICG 2271 were significantly more attractive to *T. chilonis*. *C. chlorideae* was more attracted to the leaves of ICGV 86699, ICG 2271 and ICG 1697.

Bottega *et al.* (2017) found that detrimental effects of *Podisus nigrispinus* on tomato were strongly correlated with the higher densities of type I and type IV glandular leaf trichomes in the resistant plants compared to the susceptible varieties.

Conclusion

Intermediate levels of resistance traits (e.g., trichome density) are often ideal, as they provide pest management without disrupting natural enemies, creating a more sustainable and synergistic system. The plant traits that are compatible with or even enhance biological control, such as leaf domatia, trichomes and extrafloral nectaries, which shelter and sustain natural enemies, should be selected and used in breeding programmes. undomesticated crop relatives with valuable defensive and mutualistic traits that have been lost during breeding for yield should be rediscovered and modern genetic tools should be used to reintroduce them.

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